

# Efficient Reference-pulse CNC Interpolator

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**Summary** – This report presents an efficient interpolator algorithm for reference-pulse CNC controller. The developed algorithm permits higher maximum allowable feed rate by reducing the number of required evaluated terms and employing less-extensive mathematical operations. Unlike existing interpolator which utilizes true distance calculation, the proposed interpolator is using the concept of predefined closest-distance-volumes whereby if the next point on curve is known and always resided into one of the volumes, the next interpolated point is identified. Results from computer simulations and experiments show that the developed algorithm is more efficient at least 2.5 times.

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## I. Introduction

There are two types of CNC system: reference-pulse system and sampled-data system (Yang and Hong 2001). The two systems differ mainly in interpolation method and servo control loop implementation (Koren 1983). Reference-pulse CNC system utilises incremental curve interpolator together with phase-lock servo control loop while sampled-data CNC system employs sampled-data interpolator and data-driven servo control loop.

Interpolator, or sometimes called, command generation is defined as algorithm that generates real-time reference commands to servo control loop in order to produce simultaneous multi-axis motion prescribing the shape of a mathematically defined curve (Lim et. al 1992, Lo 1992, Koren 1983, Danielsson 1970). The inputs to the interpolator are coordinates between the end points, the required speed and the mathematically-defined cutting path. Interpolator computes and generates commands to the machine-axis servo control loops in real-time such that the relative motion between the cutting tool and the work-piece follows closely to the specified cutting path and speed (Lim et. al 1992). On the other hand, servo control loop, or sometimes called, command realization is defined as servo-drive control loop that controls the positions and velocities of machine tools axes in accordance with commands generated by CNC interpolators (Srinivasan and Tsao 1997).

Reference-pulse interpolator is an incremental curve generation method in which the curve is traced by series of machine-tool smallest position resolution called basic length-unit (BLU). Using reference-pulse system, the interpolator is implemented in computer while the servo control loop is implemented by independent hardware. Reference-pulse interpolator generates sequence of reference pulses as position commands to the servo control loop axes. Each pulse represents one basic length-unit (BLU) of axis motion. At the same time, the frequency of the pulse represents the speed command to the servo control loop. Using up-down counter, servo control loop compares the reference-pulses with the actual pulses from the encoder and the difference represents position error, which is utilised by the position controller to drive the axes. In contrast, sampled-data interpolator is based upon approximating curve by series of straight-line determined by specified speed and sampling time (Lim et. al 1992). Using sampled-data system, computer is used as interpolator as well as part of the servo control loop by replacing the up-down counter with a software comparator. The computer samples the actual pulses at fixed sampling time and compares with the reference positions provided by interpolator. The difference represents position error and is utilised by software position controller, which calculates and transmits to digital-to-analogue converter (DAC) at fixed sampling time, the appropriate signal value to drive the axes (Koren 1983).

The performances of CNC systems are based on three factors: maximum allowable feed rate, maximum contour error and maximum feed rate error (Lo 1992). Maximum allowable feed rate is defined as the maximum allowable speed that the work-piece is brought into contact with the cutting tool (Krar et. al 1998). Higher maximum allowable feed rate allows higher material removal rate, which reduces machining time and cost. Maximum allowable feed rate is limited by the maximum allowable feed rate command and the maximum allowable speed of the servo-drive system. On the other hand, maximum contour error is the maximum deviation of cutter location from the desired curve (Lo 1992, Koren 1997). Contour error is a source of dimensional inaccuracies of machining part. Contour error is contributed by interpolator error (position command error and chordal error) and axial error caused by imperfection of position controller inside the servo control loop of the servo drives (Ni 1997). Finally, the maximum feed rate error is the maximum difference between the actual feed rate and the desired feed rate along the contour (Lo 1992, Koren 1997). Feed rate error is caused by feed rate command error and the imperfection of speed controller of the servo control loop.

In terms of performance, sampled-data CNC system permits higher maximum allowable feed rate and smaller feed rate error than reference-pulse CNC system (Yang and Hong 2001, Huang 1992, Lim et. al 1992, Koren 1983, Koren 1979). Nevertheless, the contour error for sampled-data CNC system is larger as well as dependent on feed rate, path curvature and sampling time. As a result, sampled-data system requires complicated servo control loop (Yang and Hong 2001, Huang 1992, Lim et. al 1992, Koren 1983, Koren 1979).

Although contour error for reference-pulse system is smaller and is independent of feed rate, path curvature and sampling time, the maximum allowable feed rate is limited and the feed rate error is bigger (Yang and Hong 2001, Huang 1992, Lim et. al 1992, Koren 1983). The maximum allowable feed rate of the reference-pulse system is determined by the maximum reference-pulse frequency generated by the reference-pulse interpolator to the servo control loop. Reference-pulse interpolator is based upon iterative technique. During iteration, interpolator algorithm

is executed and appropriate reference-pulse is generated to the servo control loops (Koren 1983, Koren 1979). Clearly, the generated reference-pulse frequency is limited by the iteration speed - higher iteration speed corresponds to higher reference-pulse frequency.

Using reference-pulse system, there are two methods to increase the iteration speed: by means of utilising faster microprocessor and by using efficient interpolator algorithm (Kiritsis 1994, Lim et. al 1992, Koren 1983). Currently, the central-processing-unit (CPU) clock-speed is typically greater than 2.0 Giga-Hertz. In comparison, the early 1990s microprocessor such as Intel 386 is only 133 Mega-Hertz. As a result, by microprocessor improvement alone, the iteration speed of reference-pulse interpolator is increasing. The other way to increase iteration speed is by employing efficient-algorithm reference-pulse interpolator. Obviously, the more efficient the algorithm the less central processing unit (CPU) time is required during interpolator iteration, which results in higher iteration speed. As noted earlier, higher iteration speed will result in higher reference-pulse frequency, which will permit higher maximum allowable feed rate. The objective of this report is to present the efficient-algorithm reference-pulse CNC interpolators that will permits higher maximum allowable feed rate. In the next section, the predefined closest-distance-volume interpolator is presented together with the commonly used Newton-Raphson interpolator technique. Efficiency comparison between the two interpolators is then presented.

## II. The Predefined Closest-Distance-Volume Interpolator

Based on Figure 1, a curve is represented by a parametric equation  $r$ . Starting from an interpolated point  $P_k$ , there are 26 possible next interpolated points.

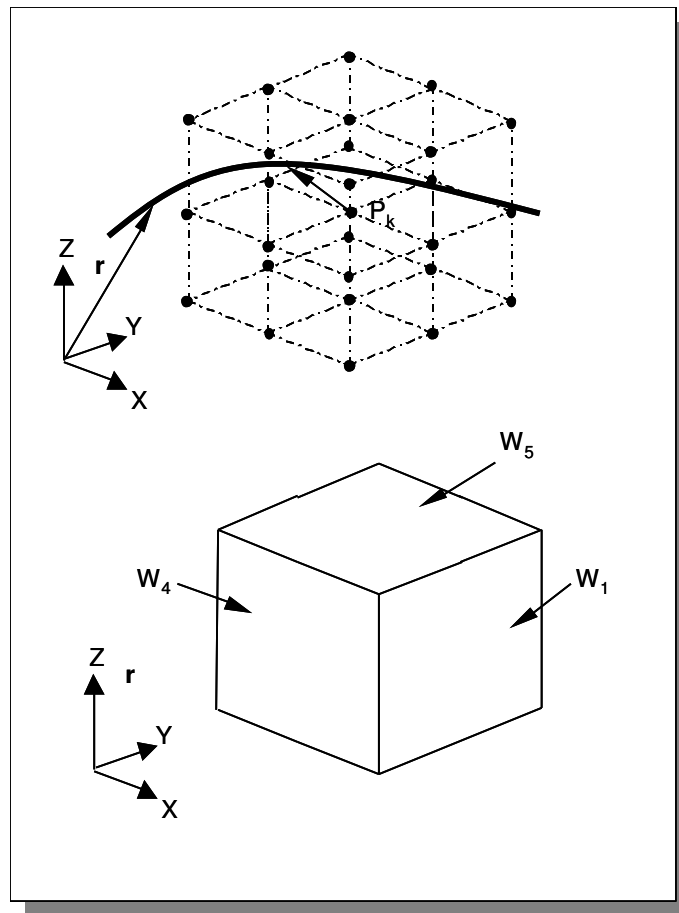


Figure 1: The 26 Possible Next Interpolated Points and the 6 Enclosing Planes

Interpolated point means a point where the motion can possibly move because each motion axis can only move in an incremental size. For instance, one possible next interpolated point is to move one step in X direction, one step in Y direction and one step in Z direction. Considering all possibilities, there are 26 possible next interpolated points. Notice that the interpolated point may not coincide with the point on curve. Thus, the interpolator problem can be described as finding the next interpolated point based the current interpolated point, the current point on the curve and the direction of the curve in order to trace the curve.

Figure 2 shows the summary of the steps for the predefined closest-distance-volume interpolator. The predefined closest-distance-volume interpolator consists of 4 major steps.

1. Determine the possible pierced plane based on tangent vector direction of the curve. In this step, the derivative of the curve at the current point on curve is calculated. From this information, the direction of the curve can be determined and thus the set of possible enclosing plane, pierced by the curve, can be determined.
2. Determine the possible pierced plane based on geometric ratio between tangent vectors of the curve. In this step, the tangent line at the point on curve is extended to the pierced plane. The geometric ratio between the X, Y and Z coordinates will indicate the set of possible enclosing planes pierced by the curve.
3. Determine the pierced plane by combining the possibility of the pierced plane set. Combining (union operation) the two sets of possible enclosing planes will uniquely determine the pierced plane. From the pierced plane, determine the next point on curve. Knowing the pierced plane, the next point on curve can be determined.
4. From the next point on curve, determine the interpolated point. Based on the next point on curve – that is sitting inside an enclosing volume (based on the pierced plane), the next interpolated point can be determined.

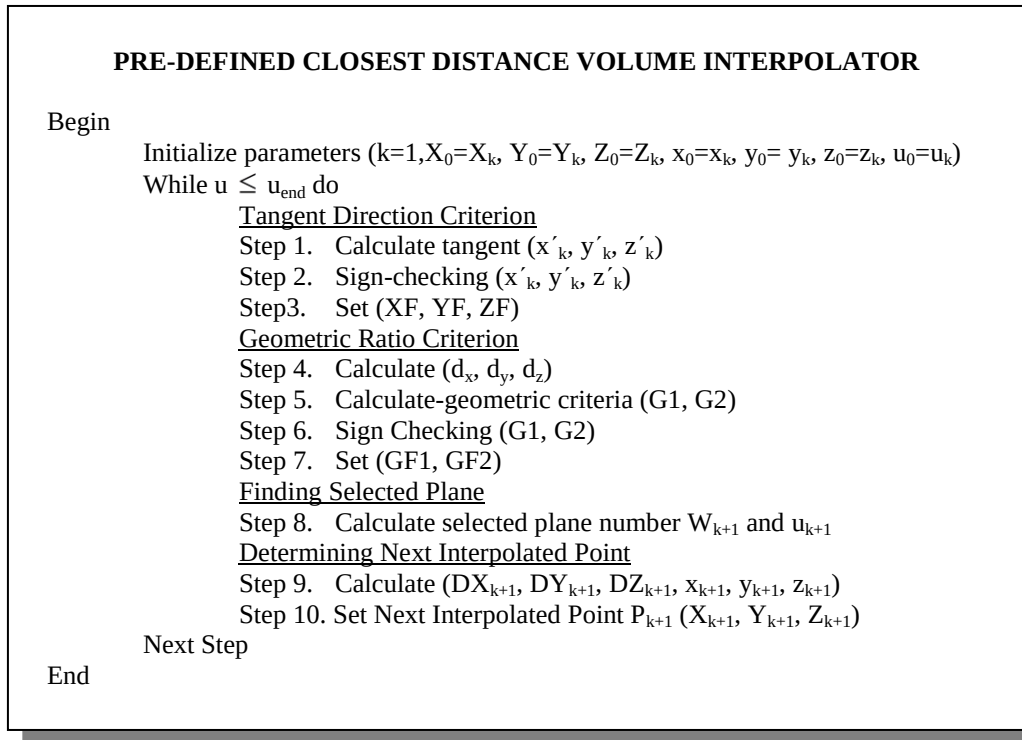


Figure 2: The predefined closest-distance-volume interpolator

For the popular Newton Raphson's interpolator, there are also 4 major steps.

1. Determine the direction of the curve based on tangent vector of the point on curve.  
This is to know where the curve is heading thus eliminating the total 26 possible points to just 7 possible points.
2. Determine the orthogonal distance from the point on curve to the all possible interpolated points.  
This distance calculation is based on Newton-Raphson method. This is a true distance – that is a straight-line distance from the point on curve to the 7 possible interpolated points.
3. Determine the shortest distance from point on curve to the interpolated points.  
From the true distance calculations, sorting algorithm is needed to determine which distance is the shortest.
4. Determine the next interpolated point based on the shortest distance.  
Clearly, the next interpolated point is the one that is the shortest distance from the point on curve.

Figure 3 shows the summary of the Newton-Raphson steps.

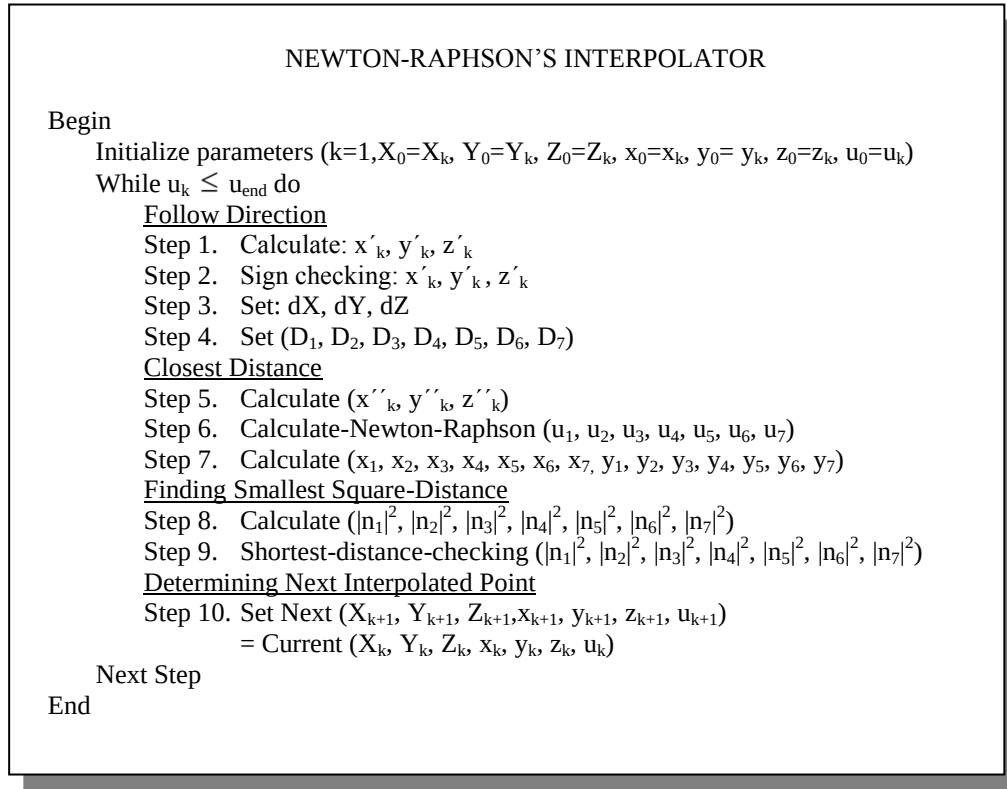


Figure 3: Newton-Raphson Interpolator

The differences between the two interpolators are the mathematical techniques used to determine the next interpolated point. The closest-distance volume technique uses little calculation but more conditional (Boolean) types operations whereas the Newton-Raphson's technique uses real computation like squaring and square rooting. This will make the Newton-Raphson technique calculation intensive. Table 1 shows the number of required terms utilised by both algorithms. It shows that Newton-Raphson technique requires 75 terms compared to only 25 terms required by the closest-distance volume algorithm.

Table 1: Required Evaluated Terms

| Interpolator | Newton-Raphson Interpolator                                                           | The Pre-defined Closest-Distance Volume Interpolator      |
|--------------|---------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Step-1       | $x'_k y'_k z'_k$                                                                      | $x'_k y'_k z'_k$                                          |
| Step-2       | -                                                                                     | -                                                         |
| Step-3       | $dX dY dZ$                                                                            | $XF YF ZF$                                                |
| Step-4       | $X_1 Y_1 Z_1 X_2 Y_2 Z_2 X_3 Y_3 Z_3 X_4 Y_4 Z_4 X_5 Y_5 Z_5 X_6 Y_6 Z_6 X_7 Y_7 Z_7$ | $d_x d_y d_z$                                             |
| Step-5       | $x''_k y''_k z''_k$                                                                   | $G1 G2$                                                   |
| Step-6       | $u_1 u_2 u_3 u_4 u_5 u_6 u_7$                                                         | -                                                         |
| Step-7       | $X_1 Y_1 Z_1 X_2 Y_2 Z_2 X_3 Y_3 Z_3 X_4 Y_4 Z_4 X_5 Y_5 Z_5 X_6 Y_6 Z_6 X_7 Y_7 Z_7$ | $GF1 GF2$                                                 |
| Step-8       | $ n_1 ^2  n_2 ^2  n_3 ^2  n_4 ^2  n_5 ^2  n_6 ^2  n_7 ^2$                             | $W_{k+1} u_{k+1}$                                         |
| Step-9       | Shortest J ShortestIndex                                                              | $X_{k+1} Y_{k+1} Z_{k+1}$<br>$DX_{k+1} DY_{k+1} DZ_{k+1}$ |
| Step-10      | $X_{k+1} Y_{k+1} Z_{k+1}$<br>$x_{k+1} y_{k+1} z_{k+1}$<br>$u_{k+1}$                   | $X_{k+1} Y_{k+1} Z_{k+1}$                                 |
| Total Terms  | 75                                                                                    | 24                                                        |

Likewise, Table 2 shows comparison of required mathematical operations between the two algorithms. It shows that Newton-Raphson's technique requires 337 operations compared to only 77 operations required by the pre-defined closest-distance volume algorithm. In particular, the square and square root operations of Newton-Raphson technique are 7 times more than the predefined closest-distance volume.

Table 2: Comparison of Required Mathematical Operations

| Mathematical Operation Types | Newton-Raphson (NR) | The Pre-defined Closest-Distance Volume (PCDV) | Difference NR - PCDV | Performance Ratio (NR/PCDV) |
|------------------------------|---------------------|------------------------------------------------|----------------------|-----------------------------|
| + and -                      | 152                 | 21                                             | 131                  | 7.24                        |
| x and ÷                      | 49                  | 5                                              | 44                   | 9.80                        |
| square and square root       | 42                  | 6                                              | 36                   | 7.00                        |
| =                            | 83                  | 24                                             | 59                   | 3.46                        |
| ≤ and ≥                      | 11                  | 21                                             | -10                  | 0.52                        |
| Total                        | 337                 | 77                                             | 260                  | 4.38                        |

### III. Simulation Results

Simulations were conducted on the two algorithms using various mathematically defined paths. Figure 4 and Figure 5 show two plots from the 2D Circle and 2D Cam profile simulations.

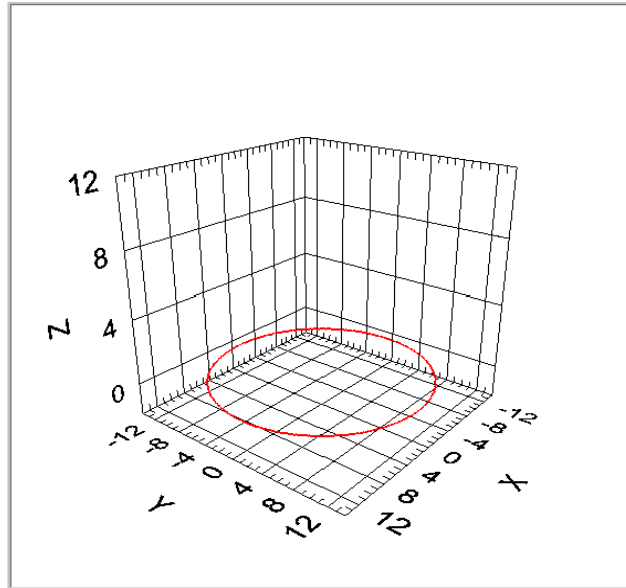


Figure 4: Simulation Plot of the 2D Circle

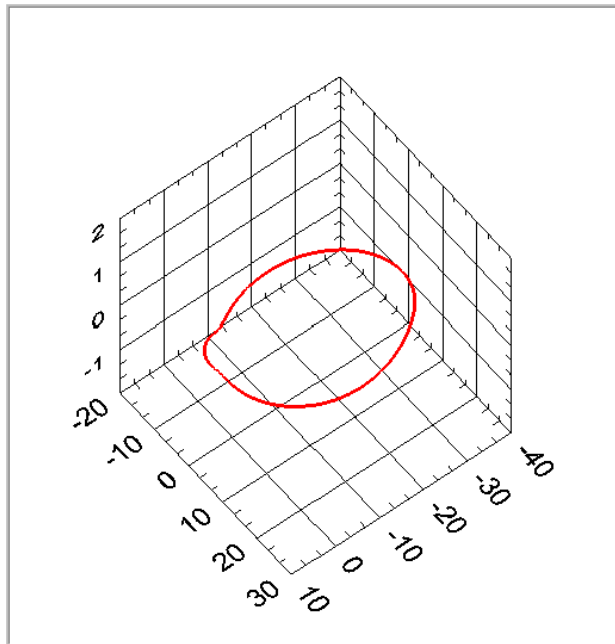


Figure III: Simulation Plot of the 2D Cam Profile

Table 3 presents the curve types and their average iteration time. Clearly, the pre-defined closest-distance volume algorithm is more efficient at least 4.47 times.

Table 3: Average Iteration Time from Computer Simulation

| Curve Type       | Newton-Raphson<br>Average Iteration Time<br>( $\mu$ Sec) | Pre-defined Closest<br>Distance Volume<br>Average Iteration Time<br>( $\mu$ Sec) | Performance Ratio<br>(NR/PCDV) |
|------------------|----------------------------------------------------------|----------------------------------------------------------------------------------|--------------------------------|
| 2D Circle        | 45.84                                                    | 9.05                                                                             | 5.07                           |
| 2D Cam profile   | 53.29                                                    | 10.50                                                                            | 5.08                           |
| 3D Straight-Line | 44.21                                                    | 9.90                                                                             | 4.47                           |
| 3D Circle        | 55.75                                                    | 10.32                                                                            | 5.40                           |
| Helix            | 48.96                                                    | 10.10                                                                            | 4.85                           |
| 3D Bezier        | 169.82                                                   | 35.88                                                                            | 4.73                           |
| 3D Polynomial    | 45.49                                                    | 8.68                                                                             | 5.24                           |

#### IV. Experiments and Results

Experimental verifications were also conducted to observe the efficiency improvement of the pre-defined closest-distance volume interpolator. Figure 6 is a picture of the experimental setup. Experiments were conducted at Control Laboratory, School of Mechanical Engineering, Universiti Sains Malaysia.



Figure 6: The Experimental Setup



Figure 7 shows the hardware setup for the experiment. A 3-axis EMCO milling machine was used as the platform to test the interpolator. The interpolator was executed in a PC-NC controller and reference pulse signals are sent to the X, Y and Z axes drives. The positions and speeds information are acquired using another PC equipped with data acquisition system.

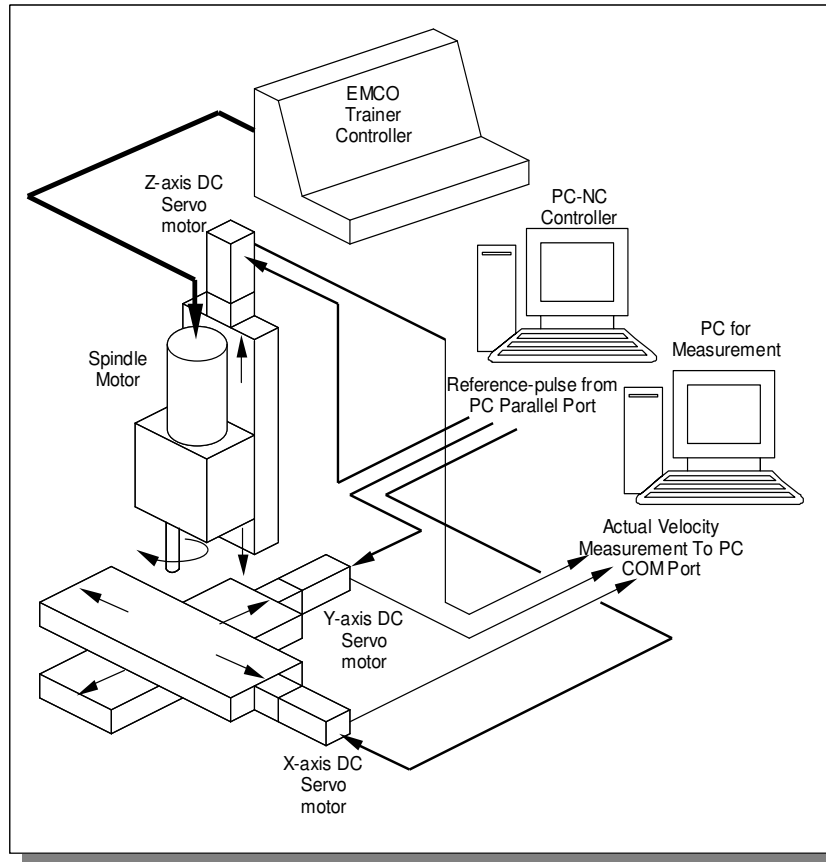


Figure 7: The Experimental Hardware Setup

Figure 8 shows the servo control loop block diagram. For each axis, a reference pulse generated by the interpolator is compared with the encoder pulse using the up-down counter. Error is then “controlled” using PID controller to drive the motor.

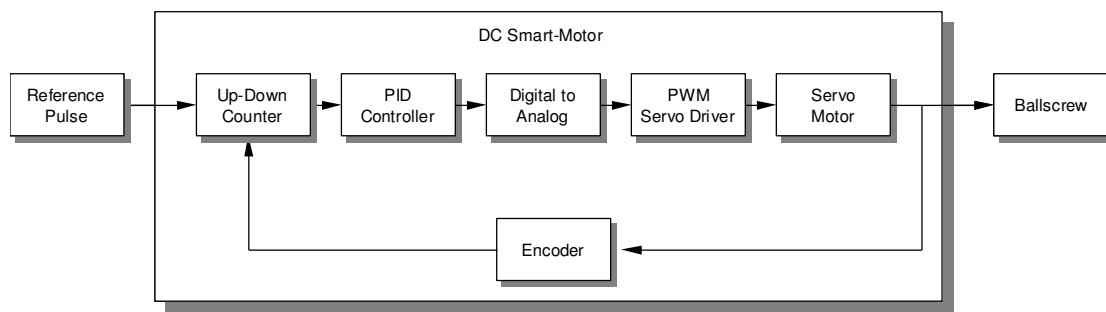


Figure 8: The Experimental Hardware Setup

The experiments were executed for two mathematical curves: 2D circle and 2D cam profile. Figure 9 shows the machining of a 2D circle while Figure 10 shows the 2D Cam profile.

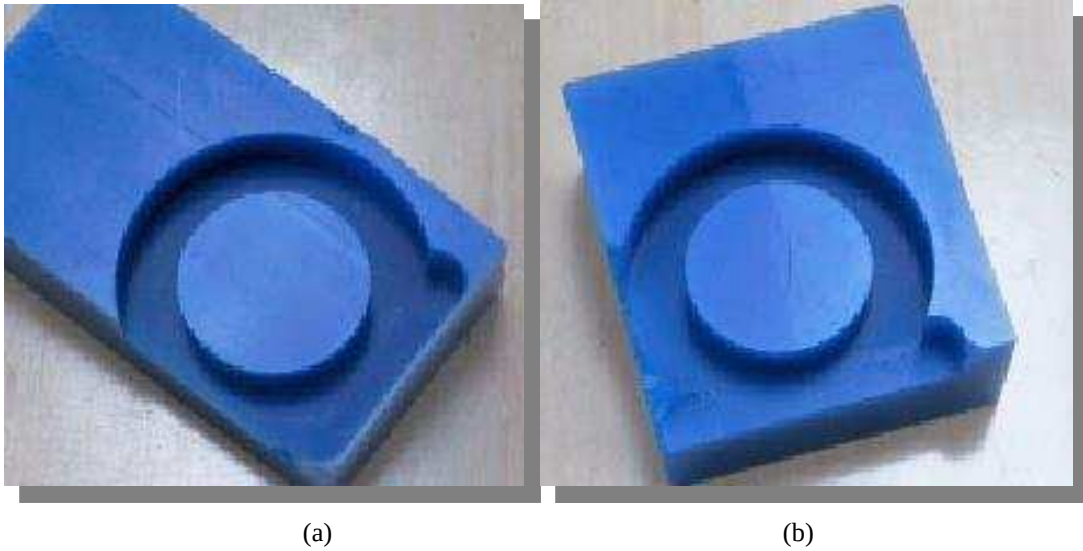


Figure 9: The machining of 2D circle using (a) the predefined closest-distance volume interpolator whereas, (b) the Newton-Raphson interpolator.



Figure 10: The machining of 2D cam profile using (a) the predefined closest-distance volume interpolator whereas, (b) the Newton-Raphson interpolator.

Figure 11 presents the results of 2D circle machining while Table 4 presents the feed rate performance between the pre-defined closest-distance volume interpolator versus the Newton-Raphson interpolator. Results indicate that the machining time for the pre-defined closest-distance volume interpolator is faster 3 times than the Newton-Raphson interpolator.

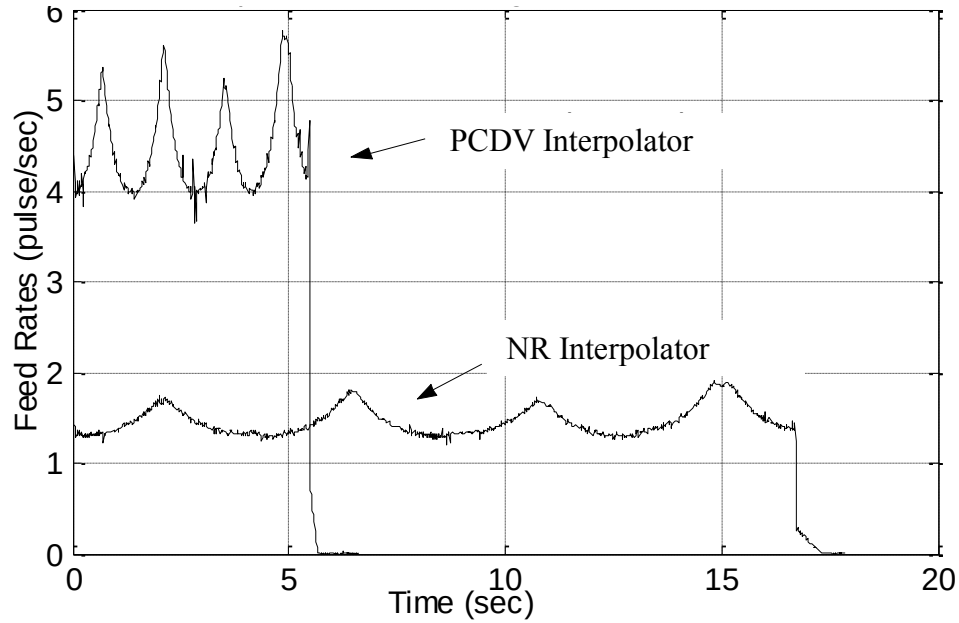


Figure 11: Feed Rates Comparison between the Pre-defined Closest Volume and the Newton-Raphson Interpolators for 2D Circle Machining

Table 4: Feed Rate Comparison for 2D Circle Machining

| Performance       | PCDV Interpolator | NR Interpolator | Performance Ratio |
|-------------------|-------------------|-----------------|-------------------|
| Maximum feed rate | 57.60 kHz         | 19.10 kHz       | 3.02              |
| Minimum feed rate | 36.37 kHz         | 12.08 kHz       | 3.01              |
| Mean feed rate    | 39.91 kHz         | 13.06 kHz       | 3.06              |
| Machining time    | 5.67 sec          | 17.32 sec       | 3.05              |

Figure 12 presents the results of 2D Cam machining while Table 5 presents the feed rates performance between the pre-defined closest-distance volume interpolator versus the Newton-Raphson interpolator. Results indicate that the machining time for the pre-defined closest-distance volume interpolator is faster at least 2.5 times than the Newton-Raphson interpolator.

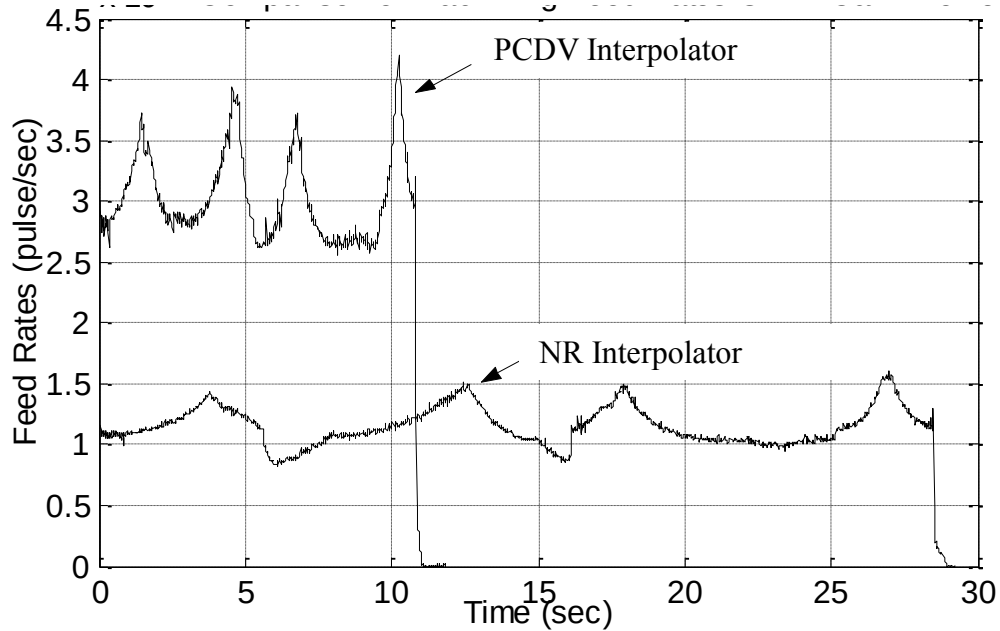


Figure 12: Feed Rates Comparison between the Pre-defined Closest Volume and the Newton-Raphson Interpolators for 2D Cam Machining

Table 5: Feed Rate Comparison for 2D Cam Machining

| Performance       | Proposed Interpolator | Kiritsis Interpolator | Performance Increase |
|-------------------|-----------------------|-----------------------|----------------------|
| Maximum feed rate | 40.11 kHz             | 16.07 kHz             | 2.50                 |
| Minimum feed rate | 26.22 kHz             | 8.28 kHz              | 3.17                 |
| Mean feed rate    | 27.24 kHz             | 10.58 kHz             | 2.57                 |
| Machining time    | 11.06 sec             | 29.11 sec             | 2.63                 |

## V. Conclusion

A pre-defined closest-distance volume interpolator was presented to increase the performance of reference-pulse CNC interpolator. It is based on conditional criteria calculation instead of calculation intensive Newton-Raphson algorithm. Results from simulations and experiments indicate that the pre-defined closest-distance volume interpolator is efficient at least 2.5 times the popular Newton-Raphson algorithm.

## References

- Danielsson, P. E. (1970). Incremental Curve Generation. *IEEE Transactions on Computers* C-19 (9): 783-793.
- Huang, J. T. (1992). Design and Application of a New CNC Command Generator for CAD/CAM Integration. *PhD Thesis, University of California, Los Angeles*.
- Kiritsis, D. (1994). High precision interpolation algorithm for 3D parametric curve generation. *Computer-Aided Design* 26(11): 850-856.
- Koren, Y. (1979). Design of Computer Control for Manufacturing Systems. *ASME Journal of Engineering for Industry* 101: 326-332.
- Koren, Y. (1983). *Computer Control of Manufacturing Systems*. New York: McGraw-Hill.
- Koren, Y. (1997). Control of Machine Tools. *ASME Journal of Manufacturing Science and Engineering* 119: 749-755.
- Krar, S. F., Rapisarda, M. and Check, A. F. (1998). *Machine Tool and Manufacturing Technology*. New York: Delmar Publishers.
- Lim, F. S., Wong, Y. S. and Rahman, M. (1992). Circular interpolators for numerical control: A comparison of the modified DDA techniques and an LSI interpolator. *Computers in Industry* 18: 41-52.
- Lo, C. C. (1992). Cross-coupling control of multi-axis manufacturing systems. *PhD Thesis, University of Michigan, Ann Arbor, Michigan*.
- Ni, J. (1997). CNC Machine Accuracy Enhancement Through Real-Time error Compensation. *ASME Journal of Manufacturing Science and Engineering* 119: 717-725.
- Srinivasan, K. and Tsao, T. C. (1997). Machine Tool Feed Drives and Their Control – A Survey of the State of the Art. *ASME Journal of Manufacturing Science and Engineering* 119: 743-748.
- Yang, Y. M. and Hong, P. W. (2001). A PC-NC milling machine with new simultaneous 3-axis control algorithm. *International Journal of Machine Tools & Manufacture* 41: 555-566.

## PUBLICATION

- Zahurin Samad and Wan Azhar Wan Yusof. Reference-Pulse Interpolator Based on Enclosing Line for 2D Parametric Curve. *Jurnal of Teknologi Part D* No 37(D) 67-78, 2002.